

Weiyi Wang
wweiyiwang@gmail.com | London, UK

EDUCATION

University College London

Sep 2025 – Sep 2026

MArch Architectural Design (RC2 – Soft Robotic Architecture)

Xi'an Jiaotong-Liverpool University & University of Liverpool

Sep 2021 – Jun 2025

BEng Architecture (Dual Degree)

- Upper Second Class Honours
- University Academic Excellence Award (Top 5%), Academic Years 2022–23 and 2023–24

RESEARCH INTERESTS

Posthuman Architecture; Interactive and Responsive Architecture; Spatial Memory; Emerging Architectural Technologies; Embodied Spatial Experience

PUBLICATIONS & CONFERENCES

Wang, W., et al. (2024). "Modular Lunar Village Concept Design for Long-Term Habitation Located in a Lava Tube." *Proceedings of the 75th International Astronautical Congress (IAC 2024)*. DOI: 10.52202/078382-0056.

RESEARCH EXPERIENCE

BAMOR — Bamboo Actuated Morphology for Operative Response

UCL Bartlett School of Architecture, RC2 – Soft Robotic Architecture

Nov 2025 – Sep 2026

- Developed a human-scale, deployable bamboo-textile structure integrating natural material behaviour, computational design and robotic actuation.
- Built a real-time interaction workflow using Python, OpenCV, MediaPipe, aubio and ROS, translating body posture and musical rhythm into adaptive structural movement.
- Explored co-embodiment through human-machine interaction, examining how responsive systems can participate in spatial experience and performance.

Modular Moon Village Concept Design for Long-term Habitation in Lava Tube

Xi'an Jiaotong-Liverpool University – Undergraduate Research Fellowship

Jun 2024 – Oct 2024

- Conducted a systematic literature review on architectural and energy strategies for extreme extraterrestrial environments.
- Developed a design-research framework translating environmental constraints into spatial systems.

Research Outcomes:

- Published paper in the 75th International Astronautical Congress (IAC 2024)
- Project presented at the International Conference on Space Habitation and Exploitation 2024
- Second Prize, Australian Space Architecture Challenge 2025

Education-led Urban Regeneration

Xi'an Jiaotong-Liverpool University – Undergraduate Research Fellowship

Jun 2023 – Sep 2023

- Reviewed and categorised literature on spatial boundaries between educational institutions and urban contexts.
- Conducted fieldwork including interviews and observational recording; analysed qualitative data and developed visual representations.
- Proposed spatial strategies informed by empirical findings.

Research Outcomes:

- The results of the research and design were further developed and won the Tiantan Academy Award

3D Printing Systems and Digital Medical Technologies

Shanghai Jiao Tong University – Summer Research Project

Jun 2025 – Aug 2025

- Reviewed additive manufacturing and soft robotic systems, focusing on material behaviour and adaptive fabrication.
- Contributed to the conceptual development of 3D-printing systems for adaptive fabrication.

SKILLS

- **Design & Visualisation:** Rhino, Grasshopper, Blender, Adobe Creative Suite
- **Computational & Interactive Prototyping:** Python, Arduino, p5.js, OpenCV, MediaPipe, ROS/ROS2, AI-assisted software prototyping
- **Research Methods:** Design Research, Literature Review, Theoretical Inquiry, Interviews, Fieldwork, Qualitative Analysis, Data Visualisation
- **Languages:** English (IELTS 7.0); Mandarin Chinese (Native)